Ideal quantum gas via grand partition function

$$Z_G = \sum_{N=0}^{\infty} \sum_{\substack{\{n_p\} \\ \sum_p n_p = N}} e^{-\frac{E(n_p)-\mu N}{\tau}} =$$

$$= \sum_{\{n_p\}} e^{-\frac{\sum_p \varepsilon_p n_p - \mu n_p}{\tau}} =$$

$$= \prod_p \sum_{n_p} e^{-\frac{(\varepsilon_p - \mu) n_p}{\tau}}$$

Here $\sum_{\{n_p\}} \ldots = \prod_p \sum_{n_p} \ldots$ is the sum over all occupation numbers n_p, for each value of momentum p.

Sum over (p) also includes sum over other quantum number (spin M_s, for example) where necessary.

In the first line, I wrote definition of Z_G — for each value of N sum over all occupation numbers such that $\sum_p n_p = N$. But since we then sum over all N, we can go to a 2nd line. There we also use that gas is <u>ideal</u>: energy of any $\{n_p\}$ distribution

$$E(\{n_p\}) = \sum_p \varepsilon_p n_p \text{ , where } \varepsilon_p = \frac{\vec{p}^{\,2}}{2m} \text{ (for example).}$$

Third line reduces Z_G to product over each allowed p-value.

Bosons: $n_p = 0, 1, 2, 3 \ldots$ Fermions: $n_p = 0, 1$

Thus for <u>bosons</u> $\sum_{n_p=0,1,2\ldots} \exp\left(-\frac{(\varepsilon_p-\mu)n_p}{\tau}\right) =$

$$= 1 + e^{-\frac{\varepsilon_p-\mu}{\tau}} + \left(e^{-\frac{\varepsilon_p-\mu}{\tau}}\right)^2 + \ldots = \frac{1}{1-e^{-\frac{\varepsilon_p-\mu}{\tau}}} \quad \left(\text{geometric series}\right)$$

for fermions $\sum_{n_p=0,1} \exp\left(-\frac{(\varepsilon_p-\mu)n_p}{\tau}\right) = 1 + e^{-\frac{\varepsilon_p-\mu}{\tau}}$.

By def: $Z_G = e^{-\Omega/\tau}$, Ω = grand (thermodynamic) potential.

$$\Rightarrow \Omega = -\tau \ln Z_G \qquad , \Omega = U - \tau\sigma - \mu N$$

$$d\Omega = -p\,dV - \sigma\,d\tau - N\,d\mu$$

$$\Rightarrow \left.\begin{aligned} \left(\frac{\partial\Omega}{\partial V}\right)_{\tau,\mu} &= -p \\ \left(\frac{\partial\Omega}{\partial \tau}\right)_{V,\mu} &= -\sigma \\ \left(\frac{\partial\Omega}{\partial \mu}\right)_{V,\tau} &= -N \end{aligned}\right\} \text{grand canonical ensemble}$$

Recall canonical also:

$$\left(\frac{\partial F}{\partial V}\right)_{\tau,N} = -p$$

$$\left(\frac{\partial F}{\partial \tau}\right)_{V,N} = -\sigma$$

$$\left(\frac{\partial F}{\partial N}\right)_{\tau,V} = \mu$$

Bosons: $\Omega_B = -\tau \sum_p \ln \frac{1}{1-e^{-\frac{\varepsilon_p-\mu}{\tau}}} = \tau \sum_p \ln\left(1-e^{-\frac{\varepsilon_p-\mu}{\tau}}\right)$

Fermions: $\Omega_F = -\tau \sum_p \ln\left(1+e^{-\frac{\varepsilon_p-\mu}{\tau}}\right)$.

Observe that we can write both cases in a single expression:

$$\Omega_{B/F} = \pm \tau \sum_p \ln\left(1 \mp e^{-\frac{\varepsilon_p - \mu}{\tau}}\right).$$

$$N = -\left(\frac{\partial \Omega}{\partial \mu}\right)_{V,\tau} = (1) \pm \tau \sum_p \frac{\mp e^{-\frac{\varepsilon_p-\mu}{\tau}}\left(\frac{1}{\tau}\right)}{1 \mp e^{-\frac{\varepsilon_p-\mu}{\tau}}} =$$

$$= \sum_p \frac{e^{-\frac{\varepsilon_p-\mu}{\tau}}}{1 \mp e^{-\frac{\varepsilon_p-\mu}{\tau}}} = \sum_p \frac{e^{\frac{\mu-\varepsilon_p}{\tau}}}{1 \mp e^{\frac{\mu-\varepsilon_p}{\tau}}} =$$

$$= \sum_p \frac{e^{\frac{\mu-\varepsilon_p}{\tau}}}{e^{\frac{\mu-\varepsilon_p}{\tau}}\left(e^{\frac{\varepsilon_p-\mu}{\tau}} \mp 1\right)} = \sum_p n(\varepsilon_p)$$

where $n(\varepsilon_p) = \dfrac{1}{e^{\frac{\varepsilon_p-\mu}{\tau}} \mp 1}$ $\quad$ ((−1) for bosons, (+1) for fermions!)

Now observe that for a uniform system:

$$\sum_p \ldots = \frac{V}{(2\pi\hbar)^3} \int d^3\vec{p} \ldots$$

$\Rightarrow$ ideal gas of bosons/fermions

$$\Omega_{B/F} = \pm \tau \frac{V}{(2\pi\hbar)^3} \int d^3\vec{p} \ln\left(1 \mp e^{-\frac{\varepsilon_p-\mu}{\tau}}\right)$$

Introduce density of states: $d^3\vec{p} \ldots = 4\pi p^2 dp \ldots$

$$\varepsilon_p = \frac{p^2}{2m} \Rightarrow dp = d\left(\sqrt{2m\varepsilon}\right) = \frac{1}{2}\sqrt{\frac{2m}{\varepsilon}}\, d\varepsilon$$

$$\Rightarrow p^2 dp = 2m\varepsilon \frac{\sqrt{2m}}{2\sqrt{\varepsilon}} d\varepsilon = \frac{1}{2}(2m)^{3/2}\sqrt{\varepsilon}\, d\varepsilon$$

$$\Rightarrow \Omega_{B/F} = \pm \tau \frac{V}{(2\pi\hbar)^3} 4\pi \cdot \frac{1}{2}(2m)^{3/2} \int d\varepsilon \sqrt{\varepsilon} \ln\left(1 \mp e^{\frac{\varepsilon-\mu}{\tau}}\right)$$

$$\Rightarrow \Omega_{B/F} = \pm \tau \int d\varepsilon\, D(\varepsilon)\, \ln\left(1 \mp e^{-\frac{\varepsilon-\mu}{\tau}}\right)$$

$$D(\varepsilon) = g \frac{V}{4\pi^2} \left(\frac{2m}{\hbar^2}\right)^{3/2} \sqrt{\varepsilon} \qquad \begin{cases} g=1 & \text{for } s=0 \text{ bosons} \\ g=2 & \text{for } s=\frac{1}{2} \text{ fermions} \end{cases}$$

$\Rightarrow$ can calculate pressure $p = -\left(\frac{\partial \Omega}{\partial V}\right)_{\tau,\mu} = -\frac{1}{V}\Omega_{B/F}$

as volume V comes in only via $D(\varepsilon)$.

$$\Rightarrow \boxed{PV = -\Omega_{B/F}}$$

Quite generally $\int_0^\infty d\varepsilon\, \varepsilon^{1/2} \ln\left(1 \mp e^{-\frac{\varepsilon-\mu}{\tau}}\right) =$

$$= \int_0^\infty \frac{2}{3} d\left(\varepsilon^{3/2}\right) \ln\left(1 \mp e^{-\frac{\varepsilon-\mu}{\tau}}\right) = -\frac{2}{3} \int_0^\infty \frac{d}{d\varepsilon}\left[\ln\left(1 \mp e^{-\frac{\varepsilon-\mu}{\tau}}\right)\right] \varepsilon^{3/2} d\varepsilon$$

Now, $\frac{d}{d\varepsilon} \ln\left(1 \mp e^{-\frac{\varepsilon-\mu}{\tau}}\right) = \frac{(\mp)\, e^{\frac{\mu-\varepsilon}{\tau}}\left(-\frac{1}{\tau}\right)}{1 \mp e^{\frac{\mu-\varepsilon}{\tau}}} =$

$$= \frac{(\pm)\, \frac{1}{\tau}\, e^{\frac{\mu-\varepsilon}{\tau}}}{e^{\frac{\mu-\varepsilon}{\tau}}\left(e^{\frac{\varepsilon-\mu}{\tau}} \mp 1\right)} = \frac{1}{\tau} \frac{\pm 1}{e^{\frac{\varepsilon-\mu}{\tau}} \mp 1}$$

$$\Rightarrow \Omega_{B/F} = \pm \tau \frac{gV}{4\pi^2} \left(\frac{2m}{\hbar^2}\right)^{3/2} \left(-\frac{2}{3}\right)\left(\frac{\pm 1}{\tau}\right) \int_0^\infty d\varepsilon\, \varepsilon^{3/2} \frac{1}{e^{\frac{\varepsilon-\mu}{\tau}} \mp 1}$$

$$= -\frac{2}{3} \frac{gV}{4\pi^2} \left(\frac{2m}{\hbar^2}\right)^{3/2} \int_0^\infty d\varepsilon\, \frac{\varepsilon^{3/2}}{e^{\frac{\varepsilon-\mu}{\tau}} \mp 1}$$

Compare with $U_{B/F} = \sum_p \varepsilon_p n_p = \frac{V}{(2\pi\hbar)^3} \int d^3\vec{p}\, \frac{\varepsilon_p}{e^{\frac{\varepsilon_p-\mu}{\tau}} \mp 1} =$

$$= \frac{gV}{4\pi^2} \left(\frac{2m}{\hbar^2}\right)^{3/2} \int_0^\infty d\varepsilon\, \frac{\varepsilon^{3/2}}{e^{\frac{\varepsilon-\mu}{\tau}} \mp 1} = -\frac{3}{2}\Omega_{B/F} = \frac{3}{2} PV$$

$\Rightarrow PV = \frac{2}{3} U$ (as for the ideal classical gas).

$$\Omega_{B/F} = -\frac{2}{3} \underbrace{\frac{gV}{4\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}}_{\text{for brevity} \to R} \int_0^\infty d\varepsilon \frac{\varepsilon^{3/2}}{e^{\frac{\varepsilon-\mu}{\tau}} \mp 1} \qquad \begin{pmatrix} -1 \text{ bosons} \\ +1 \text{ fermions} \end{pmatrix}$$

As before, define $\lambda = e^{\mu/\tau}$: $\frac{1}{e^{\frac{\varepsilon-\mu}{\tau}} \mp 1} = \frac{1}{\lambda^{-1} e^{\frac{\varepsilon}{\tau}} \mp 1} =$

$$= \frac{1}{\lambda^{-1} e^{\frac{\varepsilon}{\tau}} \left(1 \mp \lambda e^{-\frac{\varepsilon}{\tau}}\right)} = \frac{\lambda e^{-\varepsilon/\tau}}{1 \mp \lambda e^{-\varepsilon/\tau}}$$

$\leftarrow$ convenient for studying classical limit $\lambda = e^{\mu/\tau} \ll 1$.

Then for $\lambda \ll 1$: $\frac{\lambda e^{-\varepsilon/\tau}}{1 \mp \lambda e^{-\varepsilon/\tau}} \approx \lambda e^{-\varepsilon/\tau}\left(1 \pm \lambda e^{-\varepsilon/\tau} + \dots\right)$

$$\Rightarrow \Omega_{B/F} \cong -\frac{2}{3}(R) \int_0^\infty d\varepsilon\, \varepsilon^{3/2} \lambda e^{-\varepsilon/\tau}\left(1 \pm \lambda e^{-\varepsilon/\tau} + \dots\right)$$

let $x = \frac{\varepsilon}{\tau} \Rightarrow \begin{cases} d\varepsilon\, \varepsilon^{3/2} = dx\, x^{3/2} \tau^{5/2} \\ e^{-\varepsilon/\tau} = e^{-x} \end{cases}$

$$\Omega_{B/F} = -\frac{2}{3}(R)\, \tau^{5/2} \int_0^\infty dx\, x^{3/2} \lambda e^{-x} \left(1 \pm \lambda e^{-x} + \dots\right)$$

Strict Classical limit obtains when we keep only first power of $\underline{\underline{\lambda}}$

Then
$$\Omega_{class} = -\frac{2}{3}(R)\, \tau^{5/2} \int_0^\infty dx\, x^{3/2} \lambda e^{-x}$$

But $\int_0^\infty dx\, x^{3/2} e^{-x} = \frac{3}{4}\sqrt{\pi}$

$$\Rightarrow \Omega_{class} = -\frac{\sqrt{\pi}}{2} \frac{gV}{4\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} \tau^{5/2} \lambda .$$

Observe that the prefactor can be written as

$$\frac{gV}{2^3\pi^{3/2}}\left(\frac{2m}{\hbar^2}\right)^{3/2}\tau^{5/2} = gV\left(\frac{m\tau}{2\pi\hbar^2}\right)^{3/2}\tau$$

$$\Rightarrow \Omega_{class} = -gV\left(\frac{m\tau}{2\pi\hbar^2}\right)^{3/2}\tau\,\lambda$$

$$+N = -\left(\frac{\partial\Omega}{\partial\mu}\right)_{V,\tau} = gV\left(\frac{m\tau}{2\pi\hbar^2}\right)^{3/2}\tau\,\frac{\partial}{\partial\mu}\Big(\underbrace{e^{\mu/\tau}}_{\lambda}\Big) =$$

$$= gV\left(\frac{m\tau}{2\pi\hbar^2}\right)^{3/2} e^{\mu/\tau}$$

$$\Rightarrow \frac{\mu}{\tau} = \ln\left[\frac{N}{gV}\left(\frac{2\pi\hbar^2}{m\tau}\right)^{3/2}\right] = \ln\left[\frac{1}{g}\,\frac{n}{n_q}\right]$$

Recall $n_q = \left(\frac{m\tau}{2\pi\hbar^2}\right)^{3/2}$ is quantum concentration.

Thus we see that $\lambda = e^{\mu/\tau} = \frac{1}{g}\frac{n}{n_q} \ll 1$ corresponds to dilute condition $n \ll n_q$.

Let us analyze quantum corrections now:

again $$\Omega_{B/F} = -\frac{2}{3}R\,\tau^{5/2}\int_0^\infty dx\, x^{3/2}e^{-x}\,\lambda\left(1 \pm e^{-x}\lambda + \dots\right)$$

$$= -\frac{2}{3}R\,\tau^{5/2}\left\{\lambda\underbrace{\int_0^\infty dx\, x^{3/2}e^{-x}}_{\frac{3}{4}\sqrt{\pi}} \pm \lambda^2\underbrace{\int_0^\infty dx\, x^{3/2}e^{-2x}}_{\frac{1}{2^{5/2}}\frac{3}{4}\sqrt{\pi}} + \dots\right\}$$

$$= -\frac{2}{3}R\,\tau^{5/2}\,\frac{3\sqrt{\pi}}{4}\left\{\lambda \pm \frac{\lambda^2}{2^{5/2}} + \dots\right\} = -gV\,n_q\,\tau\,\lambda\left(1 \pm \frac{\lambda}{2^{5/2}}\right)$$

Hence $N = -\left(\frac{\partial \Omega}{\partial \mu}\right)_{V,T} = g V n_q T \frac{\partial}{\partial \mu}\left(\lambda \pm \frac{\lambda^2}{2^{5/2}}\right)$

But $\frac{\partial}{\partial \mu}\left(\lambda \pm \frac{\lambda^2}{2^{5/2}}\right) = \frac{\partial \lambda}{\partial \mu} \pm \frac{2\lambda}{2^{5/2}} \frac{\partial \lambda}{\partial \mu} =$

$$= \left(1 \pm \frac{\lambda}{2^{5/2}}\right) \frac{1}{T} e^{\mu/T} = \frac{1}{T}\left(1 \pm \frac{\lambda}{2^{3/2}}\right)\lambda.$$

$$\Rightarrow \quad N = g V n_q \left(1 \pm \frac{\lambda}{2^{3/2}}\right)\lambda .$$

Solve by iterations considering addition as a small correction:

$$\lambda\left(1 \pm \frac{\lambda}{2^{3/2}}\right) = \frac{N}{g V n_q} = \frac{n}{g n_q}$$

$$\Rightarrow \lambda = \frac{n}{g n_q}\left(1 \mp \frac{\lambda}{2^{3/2}}\right) = \frac{n}{g n_q}\left(1 \mp \frac{n}{2^{3/2} g n_q}\right)$$

Thus $\Omega_{B/F} = -g V n_q T \lambda \left(1 \pm \frac{\lambda}{2^{5/2}}\right) =$

$$= \begin{pmatrix}\text{plug-in} \\ \text{expression for } \lambda\end{pmatrix} = -g V n_q T \frac{n}{g n_q}\left(1 \mp \frac{n}{2^{3/2} g n_q}\right)\left(1 \pm \frac{n}{2^{5/2} g n_q}\right)$$

$$= -V n T \left(1 \mp \frac{n}{2^{3/2} g n_q} \pm \frac{n}{2^{5/2} g n_q} - \frac{n^2}{2^4 g^2 n_q^2}\right) =$$

$$\Omega_{B/F} = -N T \left\{1 \mp \frac{n}{g n_q}\underbrace{\left[\frac{1}{2^{3/2}} - \frac{1}{2^{5/2}}\right]}_{\frac{1}{2^{5/2}}} - \underbrace{\left(\frac{n}{4 g n_q}\right)^2}_{\text{neglect as } \frac{n}{n_q} \ll 1!}\right\}$$

Since $PV = -\Omega_{B/F}$, we have derived first quantum correction to classical equation of state:

$$PV = N\tau \left\{ 1 \mp \frac{1}{2^{5/2}} \frac{n}{g\, n_q} + O\left(\frac{n}{n_q}\right)^2 \right\}.$$

Upper sign $\ominus$: bosons $\rightarrow$ pressure <u>is reduced</u>

Lower sign $\oplus$: fermions $\rightarrow$ pressure <u>is increased</u>

This expression is derived assuming $\frac{n}{n_q} \ll 1$, which (for a given concentration n) means high-temperature limit.

Fully quantum behavior takes place at low τ, when $\mu_{bosons} \rightarrow 0$ at τ_{BEC}

while $\mu_{Fermions} \rightarrow \varepsilon_F \gg \tau$.

Note that $\frac{n}{n_q} = \frac{N}{V}\left(\frac{2\pi\hbar^2}{m\tau}\right)^{3/2} \sim (\hbar)^3$

Thus correction is indeed quantum !